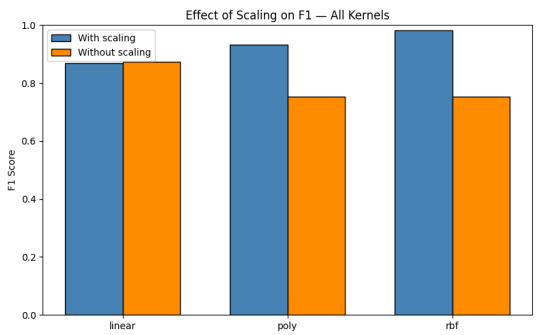
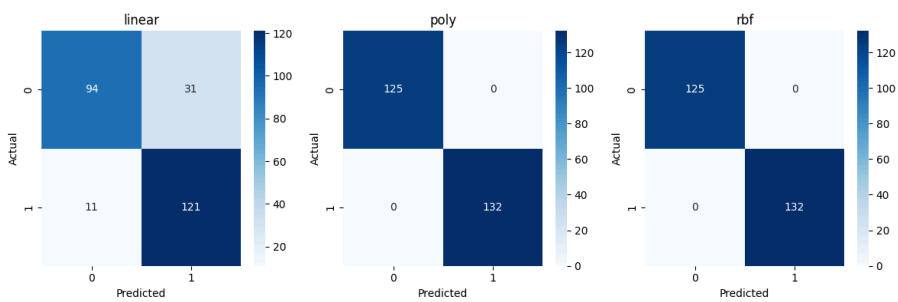
CSCI 1851 Homework 3 Writeup

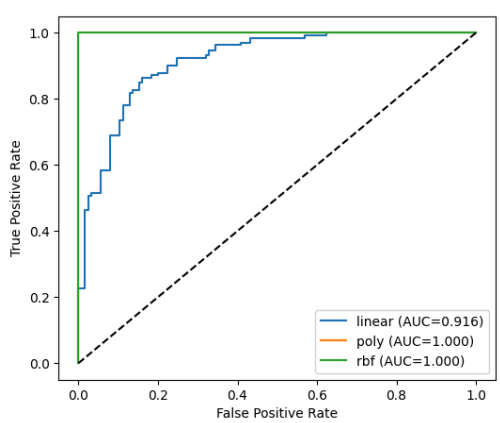
For the heart dataset, there are 1025 rows with 13 features related to patient or heart information, along with a binary value that indicates if heart disease is present or absent. There are no missing values in this dataset, and the classes are pretty balanced, with 51.3% present and 48.7% absent (526:499). Features were not highly correlated. The aging dataset has 37 sample rows with 485,516 features. Due to its small sample size, I made sure to do cross-validation for more precise results. There were 526 missing values, which I used column-wise mean imputation to address, except for a column cg01550828 which was entirely nan values, so I dropped it. I also used PCA on the aging dataset due to its large size. Both datasets were scaled using StandardScaler, which is important in SVMs because the margin is measured in the feature space, and so a larger feature value can completely dominate (ex. cholesterol has values in the 300s versus cp with 0-3), distorting the margin calculation and kernels. This showed an immediate improvement in performance (pre hyperparameter tuning) for the poly and RBF kernels on the heart data.



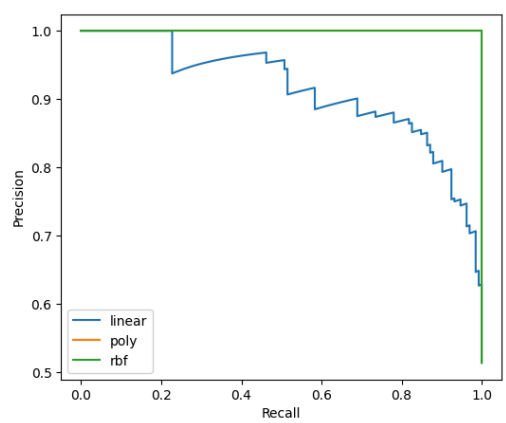
After tuning (described later) the polynomial and RBF kernels both performed perfectly with accuracy= 1.0, F1=1.0, and ROC-AUC=1.0. Linear had accuracy=0.84, F1= 0.83, and ROC-AUC =91. Poly and RBF had 0 false negatives and positives, while linear had 31 false negatives and 11 false positives.



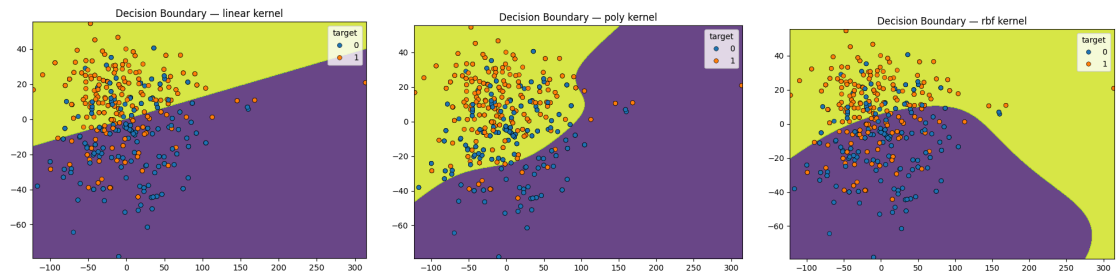
ROC Plot:



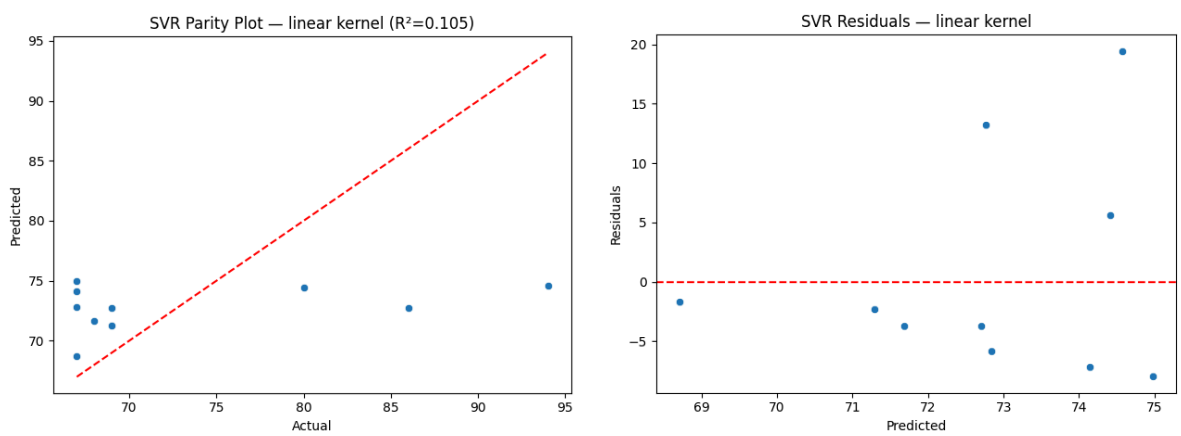
Precision-recall



This makes sense because heart disease is likely determined by non-linear interactions between features, as many of the health variables are interdependent. We can see the strong performance from the poly and RBF kernel, with precision staying high even as recall increases. The linear kernel is the weakest of the 3, confirming that decision boundary is not linearly separable, but still performs decently. We can visualize the different decision boundaries below, although it's a 2D PCA projections, so it looks like some points are being misclassified even though the scores indicate they are not:



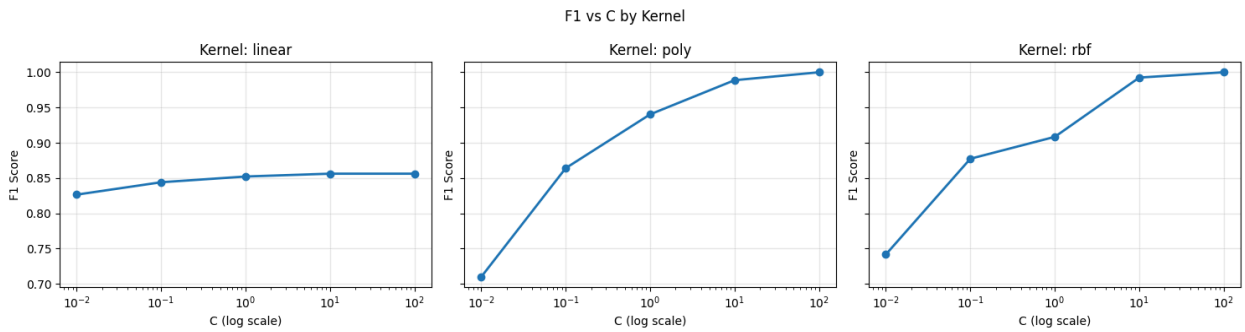
For the aging dataset, linear was the strongest, but performance was weak all around: the linear kernel with C=0.1 had an R^2 of 0.113, MAE of 7.06 years, and RMSE of 8.76 years. Every polynomial or RBF kernel had a negative R^2, suggesting it fits worse than the mean. This is likely due to the dataset constraints—with only 37 samples but very high-dimensional data, each fold trains SVR on ~30 samples, which is difficult to detect meaningful patterns from. Linear likely performs the best because it is simpler and less prone to overfitting the noise. This is reflected in the parity and residual plots below:

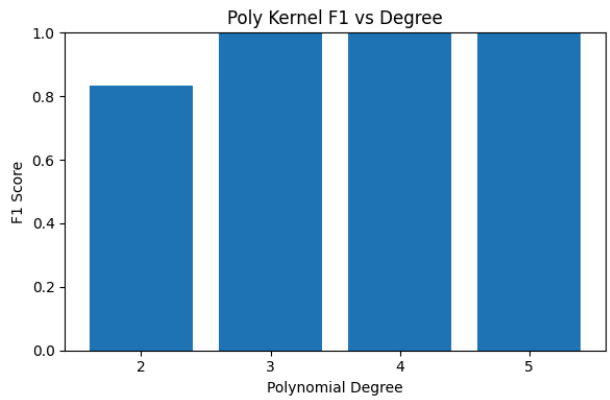


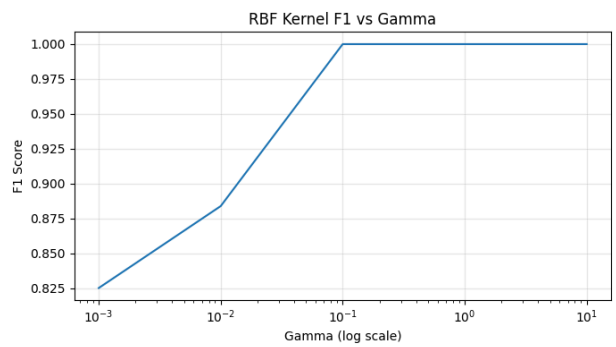
Instead of falling along the red line, predictions cluster around 70-75 regardless of actual age, suggesting the model just predicts near the mean—this explains the poor R^2 score. On the residual plot, there are large positive residuals at high actual ages, so the model underpredicts older patients. A well-fitted model would cluster randomly around 0, so this indicates the model hasn't learned the true signals of age.

A SVM finds a hyperplane that maximizes the margin between two classes. In tuning hyperparameters, I looped through three different kernels: linear, polynomial, and RBF. Kernels implicitly map data into a higher-dimensional space where a linear separator exists. For a linear kernel, that plane is straight. A polynomial kernel can capture polynomial feature interactions, with degree d controlling complexity. The RBF kernel measures similarity by distance, with y controlling the influence radius of each point. The regularization parameter C controls the width of the margin. This leads to the tradeoff between margin and slack; a large C has a narrow margin that is more prone to overfitting, while a smaller C allows for more mistakes in order for a more generalizable margin.

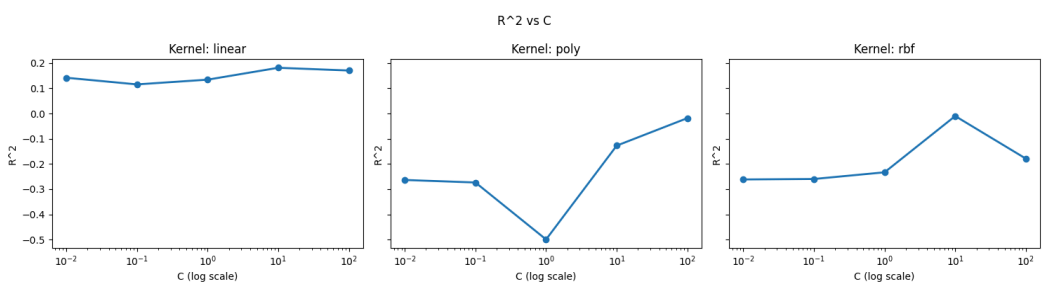
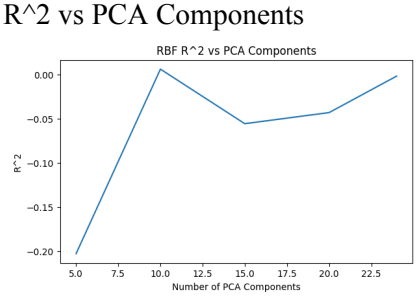
For the heart dataset, I tried out different C, degree, and gamma values:.







Initially, the polynomial and RBF kernels had ~95% accuracy. Once I updated C to 100 and degree to 3 and gamma to 1, as shown in the chart, performance for both reached accuracy, F1, and ROC-AUC of 1.0. For the aging dataset, changing the hyperparameters didn't greatly improve performance for the poly and RBF kernels. They still were negative even after tuning, and changing C slightly impacted the linear kernel, with optimal C = 10. Finally, I tuned the PCA components, finding that the optimal was 10.



R^2 vs PCA Components